Vectors in n Dimensions

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Vector Arithmetic and Linear Combinations**

Consider two vectors $\mathbf{v}_1 = (3; 4)$ and $\mathbf{v}_2 = (-4; 3)$. What is the dot product $\mathbf{v}_1 \cdot \mathbf{v}_2$?

The dot product of two vectors $\mathbf{v}_1 = (a; b)$ and $\mathbf{v}_2 = (c; d)$ is given by $\mathbf{v}_1 \cdot \mathbf{v}_2 = ac + bd$. In this case, $\mathbf{v}_1 = (3; 4)$ and

2 **[Foundational]** • **Target: Linear Independence and Dependence**

Consider the vectors $\mathbf{v}_1 = (3; 4)$ and $\mathbf{v}_2 = (-4; 3)$. What is the dot product $\mathbf{v}_1 \cdot \mathbf{v}_2$?

To find the dot product of two vectors $\mathbf{v}_1 = (a; b)$ and $\mathbf{v}_2 = (c; d)$, we compute $\mathbf{v}_1 \cdot \mathbf{v}_2 = ac + bd$. In this case, $\mathbf{v}_1 = (3; 4)$ and $\mathbf{v}_2 = (-4; 3)$. Therefore, the dot product is $(3)(-4) + (4)(3) = -12 + 12 = 0$. Since the dot product is 0, the vectors are orthogonal.

3 **[Foundational]** • **Target: Dot Product, Norm, and Orthogonality**

Express the complete solution set of the non-homogeneous linear system in parametric vector form:
$\{x_1 + 2x_2 - x_3 + 3x_4 = 4; 2x_1 + 4x_2 - x_3 + 8x_4 = 11; -x_1 - 2x_2 + 2x_3 - x_4 = -1$

Step 1: Set up augmented matrix and row reduce:
$(1, 2, -1, 3, 4; 2, 4, -1, 8, 11; -1, -2, 2, -1, -1) \rightarrow (1, 2, -1, 3, 4; 0, 0, 1, 2, 3; 0, 0, 1, 2, 3)$
$\rightarrow (1, 2, 0, 5, 7; 0, 0, 1, 2, 3; 0, 0, 0, 0, 0)$. Step 2: Identify pivot and free variables: Pivots in columns 1 and 3 (x_1, x_3). Free variables are $x_2 = s$, $x_4 = t$. From RREF: $x_1 = 7 - 2s - 5t$, $x_3 = 3 - 2t$. Step 3: Parametric vector decomposition.

4 **[Foundational]** • **Target: Orthogonal Projections**

Consider the vector $\mathbf{v} = (-3; 4)$ in $\mathbb{R}^2$. What is the magnitude (or norm) of this vector?

The magnitude (or norm) of a vector $\mathbf{v} = (x; y)$ in $\mathbb{R}^2$ is given by the formula $= x^2 + y^2$. For the given vector $\mathbf{v} = (-3; 4)$, we have $x = -3$ and $y = 4$. Therefore, the magnitude is
$= (-3)^2 + (4)^2 = 9 + 16 = 25 = 5$.

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5 [Foundational] • Target: Vector Arithmetic and Linear Combinations

Consider the vector $\mathbf{a} = (2\ k)$ and the vector $\mathbf{b} = (-1\ 3)$. \\ Part 1: If $\mathbf{a}$ and $\mathbf{b}$ are orthogonal, find the value of k. \\ Part 2: Let the resulting vector from Part 1 be $\mathbf{v}$. The transformation T is defined by the matrix $M = (1, 2\ 2, -1)$. \\ If $T(\mathbf{v}) = (x\ y)$, find the value of x.

Part 1: For two vectors to be orthogonal, their dot product must be zero. \\ $\mathbf{a} \cdot \mathbf{b} = (2)(-1) + (k)(3) = -2 + 3k$. \\ Setting the dot product to zero for orthogonality: \\ $-2 + 3k = 0 \Rightarrow 3k = 2 \Rightarrow k = \frac{2}{3}$. \\ Thus, $\mathbf{v} = \left(2\ \frac{2}{3}\right)$. \\ Part 2: The transformation T is given by the matrix $M = (1, 2\ 2, -1)$. \\ We need to find $T(\mathbf{v}) = M\mathbf{v}$. \\.

6 [Foundational] • Target: Linear Independence and Dependence

A linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ is defined by $T(x, y) = (ax + by, cx + dy)$. The matrix representation of T with respect to the standard basis is $A = (a, b\ c, d)$. The image of the standard basis vectors $\mathbf{e}_1 = (1, 0)$ and $\mathbf{e}_2 = (0, 1)$ under T are given by $T(\mathbf{e}_1) = (3, -1)$ and $T(\mathbf{e}_2) = (1, 2)$. The transformation is applied to a square with vertices $(0, 0), (1, 0), (0, 1), (1, 1)$. The area of the transformed square is known to be k. Find the value of k.

The matrix representation of the linear transformation T with respect to the standard basis is formed by the images of the standard basis vectors as its columns. Therefore, $A = (3, 1\ -1, 2)$. The area of the transformed region is the absolute value of the determinant of the matrix A multiplied by the area of the original region. The original region is a square with vertices $(0, 0), (1, 0), (0, 1), (1, 1)$, which has an area of $1 \times 1 = 1$. The determinant of A is.

7 [Foundational] • Target: Dot Product, Norm, and Orthogonality

A parallelogram in the xy-plane is defined by the vectors $\mathbf{u} = (3; 1)$ and $\mathbf{v} = (1; 4)$. What is the area of this parallelogram?

The area of a parallelogram defined by two vectors $\mathbf{u} = (u_1; u_2)$ and $\mathbf{v} = (v_1; v_2)$ in $\mathbb{R}^2$ is given by the absolute value of the determinant of the matrix formed by these vectors as columns (or rows). In this case, the matrix is $A = (3, 1; 1, 4)$. The determinant of A is $\det(A) = (3)(4) - (1)(1) = 12 - 1 = 11$. Therefore, the area of the parallelogram is $|11| = 11$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Orthogonal Projections

Consider the vector $\mathbf{v} = (3; 4)$. If this vector is plotted on a standard xy-plane, it represents a directed line segment from the origin to the point $(3, 4)$. What is the geometric interpretation of the length of this vector?

The length of a vector $\mathbf{v} = (x; y)$ is given by its magnitude, $= x^2 + y^2$. Geometrically, this corresponds to the distance from the origin $(0, 0)$ to the point (x, y) on the Cartesian plane, as described by the Pythagorean theorem. For $\mathbf{v} = (3; 4)$, the length is $\sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$. This value represents the distance of the point $(3, 4)$ from the origin.

9 [Intermediate] • Target: Vector Arithmetic and Linear Combinations

Consider the parallelepiped in $\mathbb{R}^3$ defined by the vectors $\mathbf{u} = (2, 1, -1)$, $\mathbf{v} = (1, k, 2)$, and $\mathbf{w} = (3, 4, 1)$. The volume of this parallelepiped is given as 15 cubic units. Find the possible value(s) of k.

The volume of a parallelepiped defined by three vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ in $\mathbb{R}^3$ is given by the absolute value of their scalar triple product, which is equivalent to the absolute value of the determinant of the matrix formed by these vectors as its rows (or columns). The matrix formed by the given vectors is: $M = [2, 1, -1\ 1, k, 2\ 3, 4, 1]$ We calculate the determinant of M.

10 [Intermediate] • Target: Linear Independence and Dependence

Consider a parallelepiped in $\mathbb{R}^3$ formed by the origin and the three vectors $\mathbf{u} = (2\ 1\ k)$, $\mathbf{v} = (1\ 3\ 0)$, and $\mathbf{w} = (0\ 2\ 4)$. The volume of this parallelepiped is given as 24 cubic units. Find the value of k if $k > 0$.

The volume of a parallelepiped formed by three vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ is given by the absolute value of their scalar triple product, which can be computed as the absolute value of the determinant of the matrix formed by these vectors as columns (or rows). Let the matrix be $M = (2, 1, 0\ 1, 3, 2\ k, 0, 4)$. The determinant of M is calculated as: $\det(M) = 2|3, 2\ 0, 4| - 1|1, 2\ k, 4| + 0|1, 3\ k, 0|$.

11 [Intermediate] • Target: Dot Product, Norm, and Orthogonality

Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation represented by the matrix $A = (3, -1\ 1, 2)$ with respect to the standard basis $\{\mathbf{e}_1, \mathbf{e}_2\}$. Consider a new basis $B = \{\mathbf{v}_1, \mathbf{v}_2\}$ where $\mathbf{v}_1 = (1\ 1)$ and $\mathbf{v}_2 = (1\ -1)$. Let P be the change-of-basis matrix from basis B to the standard basis. Find the matrix representation of T with respect to basis B, denoted by $[T]_B$. This matrix can be expressed as $P^{-1}AP$. Find the sum of the diagonal entries of $[T]_B$.

The change-of-basis matrix P from basis B to the standard basis has the vectors of B as its columns. Thus, $P = (1, 1\ 1, -1)$. The inverse of P is $P^{-1} = \frac{1}{\det(P)}(-1, -1\ -1, 1)$. Since $\det(P) = (1)(-1) - (1)(1) = -2$, we have $P^{-1} = \frac{1}{-2}(-1, -1\ -1, 1) = (1/2, 1/2\ 1/2, -1/2)$. The matrix representation of T with respect to basis B is given by $[T]_B = P^{-1}AP$. First, we compute AP.

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12 [Intermediate] • Target: Orthogonal Projections

Solve the initial value problem for the coupled system of linear differential equations: $\mathbf{x}'(t) = (1, 4; 2, 3)\mathbf{x}(t), \qquad \mathbf{x}(0) = (5; 1).$

Step 1: Eigenvalues of coefficient matrix: $\det(A - \lambda I) = (1 - \lambda)(3 - \lambda) - 8 = \lambda^2 - 4\lambda - 5 = (\lambda - 5)(\lambda + 1) = 0 \Rightarrow \lambda_1 = 5,\ \lambda_2 = -1.$ Step 2: Eigenvectors:For $\lambda_1 = 5$: $(A - 5I) = (-4, 4; 2, -2) \Rightarrow \mathbf{v}_1 = (1; 1)$.For $\lambda_2 = -1$: $(A + I) = (2, 4; 2, 4) \Rightarrow \mathbf{v}_2 = (-2; 1)$. Step 3: General solution and initial conditions.

13 [Intermediate] • Target: Vector Arithmetic and Linear Combinations

Consider a linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^2$ defined by $T(x, y, z) = (2x - y + z, x + 3y - 2z)$. Let V be the image of T, i.e., $V = \text{Im}(T)$. Furthermore, let W be the subspace of $\mathbb{R}^2$ defined by $W = \{(a, b) \in \mathbb{R}^2 \mid a + b = 0\}$. Find the dimension of the intersection of V and W, i.e., $\dim(V \cap W)$.

The linear transformation T maps $\mathbb{R}^3$ to $\mathbb{R}^2$. The matrix representation of T is $A = (2, -1, 1\ 1, 3, -2)$. The image of T, denoted as $V = \text{Im}(T)$, is the column space of A. The columns of A are $(2\ 1)$, $(-1\ 3)$, and $(1\ -2)$. Since these vectors span V, we can find a basis for V by finding the pivot columns of A or by noting that the image is a subspace of $\mathbb{R}^2$. The rank of A is 2, which means.

14 [Intermediate] • Target: Linear Independence and Dependence

Prove the Parallelogram Law for arbitrary vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$: $||\mathbf{u} + \mathbf{v}||^2 + ||\mathbf{u} - \mathbf{v}||^2 = 2||\mathbf{u}||^2 + 2||\mathbf{v}||^2$. What is its geometric significance?

Algebraic Proof: Expand each squared Euclidean norm using the inner product: $||\mathbf{u} + \mathbf{v}||^2 = (\mathbf{u} + \mathbf{v}) \cdot (\mathbf{u} + \mathbf{v}) = ||\mathbf{u}||^2 + 2(\mathbf{u} \cdot \mathbf{v}) + ||\mathbf{v}||^2.$ $||\mathbf{u} - \mathbf{v}||^2 = (\mathbf{u} - \mathbf{v}) \cdot (\mathbf{u} - \mathbf{v}) = ||\mathbf{u}||^2 - 2(\mathbf{u} \cdot \mathbf{v}) + ||\mathbf{v}||^2.$ Adding the two equations directly cancels the cross terms $2(\mathbf{u} \cdot \mathbf{v})$ and.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Dot Product, Norm, and Orthogonality • Cross-Lesson: 4.01 Determinants

The vector $\mathbf{v} = (-2; 1)$ is an eigenvector of the matrix $A = (3, 2; 1, 4)$. What is the corresponding eigenvalue λ?

By definition, a non-zero vector $\mathbf{v}$ is an eigenvector of A corresponding to eigenvalue λ if and only if $A\mathbf{v} = \lambda\mathbf{v}$. Compute the matrix-vector product $A\mathbf{v}$: $A\mathbf{v} = (3, 2; 1, 4)(-2; 1) = (3(-2) + 2(1); 1(-2) + 4(1)) = (-6 + 2; -2 + 4) = (-4; 2)$ Comparing $A\mathbf{v}$ to $\lambda\mathbf{v}$: $(-4; 2) = \lambda(-2; 1)$ This gives $-4 = -2\lambda$ and $2 = 1\lambda$, which.

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16 [Challenging] • Target: Orthogonal Projections • Cross-Lesson: 2.01 Systems of Linear Equations

Let $M_2(\mathbb{R})$ be the vector space of 2×2 real matrices. Determine, with formal verification via the Subspace Test, whether $W = \{A \in M_2(\mathbb{R}) : A^T = -A\}$ (the set of skew-symmetric matrices) is a subspace of $M_2(\mathbb{R})$. Find a basis and $\dim(W)$.

Subspace Test Verification:Non-emptiness: The zero matrix O satisfies $O^T = O = -O$, so $O \in W$ .Additivity: Let $A, B \in W$. Then $(A + B)^T = A^T + B^T = (-A) + (-B) = -(A + B)$. Thus $A + B \in W$.Scalar Closure: Let $A \in W$ and $c \in \mathbb{R}$. Then $(cA)^T = c(A^T) = c(-A) = -(cA)$. Thus $cA \in W$.Since all 3 axioms hold, W is a vector subspace of $M_2(\mathbb{R})$. General form and basis: A matrix $A = (a, b; c, d)$ satisfies.

17 [Challenging] • Target: Vector Arithmetic and Linear Combinations • Cross-Lesson: 3.01 Matrices & Matrix Algebra

Express the invertible matrix $A = (1, 2; 3, 4)$ as a product of elementary matrices.

Step 1: Reduce A to I_2 via row operations: 1. $R_2 \to R_2 - 3R_1$: Elementary matrix $E_1 = (1, 0; -3, 1)$. Result: $(1, 2; 0, -2)$. 2. $R_2 \to -\frac{1}{2}R_2$: Elementary matrix $E_2 = (1, 0; 0, -1/2)$. Result: $(1, 2; 0, 1)$. 3. $R_1 \to R_1 - 2R_2$: Elementary matrix $E_3 = (1, -2; 0, 1)$. Result: I_2. Step 2: Invert elementary matrices: We have $E_3E_2E_1A = I_2 \Rightarrow A = E_1^{-1}E_2^{-1}E_3^{-1}$. Computin.

18 [Challenging] • Target: Linear Independence and Dependence • Cross-Lesson: 4.01 Determinants

Consider the following system of linear equations represented by the augmented matrix: $(1, 2, |, 5; 2, -1, |, 0)$. Use row reduction to transform the matrix into row-echelon form and determine the unique solution (x, y) for the system. What is the value of x?

We are given the augmented matrix: $(1, 2, |, 5; 2, -1, |, 0)$ To transform this matrix into row-echelon form, we can perform the following row operation: $R_2 \leftarrow R_2 - 2R_1$.
$(1, 2, |, 5; 2 - 2(1), -1 - 2(2), |, 0 - 2(5))$ $(1, 2, |, 5; 0, -5, |, -10)$ Now, we can perform another row operation: $R_2 \leftarrow R_2/-5$. $(1, 2, |, 5; 0, 1, |, 2)$ This matrix is now in row-echelon form. The corresponding system of equations is: $x + 2y = 5$ $y = 2$ Substitut.

19 [Challenging] • Target: Dot Product, Norm, and Orthogonality • Cross-Lesson: 2.01 Systems of Linear Equations

Let $\mathbf{u} = (1, -2, 2, 0)^T$ and $\mathbf{v} = (2, 1, 0, -2)^T$ in $\mathbb{R}^4$. Compute $||\mathbf{u}||$, $||\mathbf{v}||$, the inner product $\mathbf{u} \cdot \mathbf{v}$, the cosine of the angle θ between them, and verify the Cauchy–Schwarz inequality.

1. Norms: $||\mathbf{u}|| = \sqrt{1^2 + (-2)^2 + 2^2 + 0^2} = \sqrt{1 + 4 + 4 + 0} = \sqrt{9} = 3.$
$||\mathbf{v}|| = \sqrt{2^2 + 1^2 + 0^2 + (-2)^2} = \sqrt{4 + 1 + 0 + 4} = \sqrt{9} = 3.$ 2. Inner Product:
$\mathbf{u} \cdot \mathbf{v} = (1)(2) + (-2)(1) + (2)(0) + (0)(-2) = 2 - 2 + 0 + 0 = 0.$ 3. Angle:
$\cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{||\mathbf{u}||||\mathbf{v}||} = \frac{0}{3 \cdot 3} = 0 \Rightarrow \theta = \frac{\pi}{2}$ rad (90°). 4. Cauchy–Schwarz Verification.

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20 [Challenging] • Target: Orthogonal Projections • Cross-Lesson: 3.01 Matrices & Matrix Algebra

Consider the linear transformation $T: \mathbb{R}^2 \to \mathbb{R}^2$ represented by the matrix $A = (3, 1; 1, 3)$. Identify one of the eigenvalues of A.

To find the eigenvalues of a matrix A, we solve the characteristic equation $\det(A - \lambda I) = 0$, where λ represents the eigenvalues and I is the identity matrix. For the given matrix $A = (3, 1; 1, 3)$, the characteristic equation is: $\det(3 - \lambda, 1; 1, 3 - \lambda) = 0$ This expands to: $(3 - \lambda)(3 - \lambda) - (1)(1) = 0$ $(3 - \lambda)^2 - 1 = 0$ $9 - 6\lambda + \lambda^2 - 1 = 0$ $\lambda^2 - 6\lambda + 8 = 0$ Factoring the quadratic equation.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Vector Arithmetic and Linear Combinations

Given $\mathbf{u} = [2, -1, 4]$ and $\mathbf{v} = [-3, 5, 2]$ in $\mathbb{R}^3$, compute $3\mathbf{u} - 2\mathbf{v}$.

a. $[12, -13, 8]$ ✓

b. $[0, 7, 16]$

c. $[12, -7, 8]$

d. $[-12, 13, -8]$

$3\mathbf{u} = [6, -3, 12]$. $-2\mathbf{v} = [6, -10, -4]$. Sum $= [6 + 6, -3 - 10, 12 - 4] = [12, -13, 8]$.

22 [Foundational] • Target: Vector Arithmetic and Linear Combinations

Is the vector $\mathbf{w} = [7, 4]$ a linear combination of $\mathbf{v}_1 = [1, 2]$ and $\mathbf{v}_2 = [3, -1]$?

a. **Yes, $\mathbf{w} = \frac{19}{7}\mathbf{v}_1 + \frac{10}{7}\mathbf{v}_2$** ✓

b. No, the vectors are linearly dependent

c. Yes, $\mathbf{w} = 2\mathbf{v}_1 + \mathbf{v}_2$

d. No, because the determinant is zero

Solve $c_1[1, 2] + c_2[3, -1] = [7, 4] \Rightarrow c_1 + 3c_2 = 7$ and $2c_1 - c_2 = 4 \Rightarrow c_2 = 2c_1 - 4$. Substitute: $c_1 + 6c_1 - 12 = 7 \Rightarrow 7c_1 = 19 \Rightarrow c_1 = 19/7, c_2 = 10/7$.

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23 [Foundational] • Target: Vector Arithmetic and Linear Combinations

What is the definition of the span of a set of vectors $\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ in $\mathbb{R}^n$?

a. The set of all possible linear combinations $c_1\mathbf{v}_1 + \ldots + c_k\mathbf{v}_k$ where $c_i \in \mathbb{R}$ ✓

b. The set of all vectors perpendicular to $\mathbf{v}_1$

c. The sum of the lengths of the vectors

d. The matrix formed by using the vectors as columns

By definition, $\text{Span}\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ is the subspace consisting of all linear combinations of those vectors.

24 [Foundational] • Target: Linear Independence and Dependence

A set of vectors $\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ is linearly independent if and only if:

a. $c_1\mathbf{v}_1 + \ldots + c_k\mathbf{v}_k = \mathbf{0}$ implies $c_1 = c_2 = \ldots = c_k = 0$ ✓

b. At least one vector is a multiple of another

c. The dot product of any two vectors is zero

d. The number of vectors k is greater than the dimension n

Linear independence requires that the only linear combination yielding the zero vector is the trivial solution where all coefficients are zero.

25 [Foundational] • Target: Linear Independence and Dependence

Can any set of 4 vectors in $\mathbb{R}^3$ be linearly independent?

a. No, any set of more than n vectors in $\mathbb{R}^n$ is guaranteed to be linearly dependent ✓

b. Yes, provided none of the vectors is the zero vector

c. Yes, if they all have unit length

d. Only if their cross products are non-zero

By the Fundamental Theorem of Linear Algebra, any collection of $k > n$ vectors in $\mathbb{R}^n$ must be linearly dependent because there are at most n pivot columns.

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26 [Foundational] • Target: Linear Independence and Dependence

Are the vectors $\mathbf{v}_1 = [1, 2, 0]$, $\mathbf{v}_2 = [0, 1, 3]$, and $\mathbf{v}_3 = [2, 5, 3]$ linearly independent?

a. No, because $\mathbf{v}_3 = 2\mathbf{v}_1 + \mathbf{v}_2$ ✓

b. Yes, they span all of $\mathbb{R}^3$

c. No, because they contain a zero entry

d. Yes, because their determinant is 1

Check: $2\mathbf{v}_1 + \mathbf{v}_2 = 2[1, 2, 0] + [0, 1, 3] = [2, 4 + 1, 3] = [2, 5, 3] = \mathbf{v}_3$. Since one vector is a combination of the others, the set is dependent.

27 [Foundational] • Target: Dot Product, Norm, and Orthogonality

Compute the Euclidean norm (length) of the vector $\mathbf{v} = [2, -3, 6]$ in $\mathbb{R}^3$.

a. 7 ✓

b. $\sqrt{11}$

c. 49

d. $\sqrt{41}$

$||\mathbf{v}|| = \sqrt{2^2 + (-3)^2 + 6^2} = \sqrt{4 + 9 + 36} = \sqrt{49} = 7$.

28 [Foundational] • Target: Dot Product, Norm, and Orthogonality

Find the unit vector in the direction of $\mathbf{u} = [-1, 2, 2]$.

a. $\left[-\frac{1}{3}, \frac{2}{3}, \frac{2}{3}\right]$ ✓

b. $[-1, 2, 2]$

c. $\left[-\frac{1}{9}, \frac{2}{9}, \frac{2}{9}\right]$

d. $\left[-\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}, 0\right]$

$||\mathbf{u}|| = \sqrt{(-1)^2 + 2^2 + 2^2} = \sqrt{1 + 4 + 4} = 3$. The unit vector is $\frac{\mathbf{u}}{||\mathbf{u}||} = [-1/3, 2/3, 2/3]$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Dot Product, Norm, and Orthogonality

What does the Cauchy-Schwarz inequality state for any vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$?

a. $|\mathbf{u} \cdot \mathbf{v}| \leq ||\mathbf{u}||||\mathbf{v}||$ ✓

b. $||\mathbf{u} + \mathbf{v}|| \leq ||\mathbf{u}|| + ||\mathbf{v}||$

c. $\mathbf{u} \cdot \mathbf{v} = ||\mathbf{u}||||\mathbf{v}||$

d. $|\mathbf{u} \cdot \mathbf{v}| \geq ||\mathbf{u}||||\mathbf{v}||$

The Cauchy-Schwarz inequality states $|\mathbf{u} \cdot \mathbf{v}| \leq ||\mathbf{u}||||\mathbf{v}||$, with equality holding if and only if the vectors are collinear.

30 [Intermediate] • Target: Orthogonal Projections

What is the formula for the orthogonal projection of vector $\mathbf{y}$ onto a non-zero vector $\mathbf{u}$?

a. $\text{proj}_{\mathbf{u}}(\mathbf{y}) = \frac{\mathbf{y} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}}\mathbf{u}$ ✓

b. $\text{proj}_{\mathbf{u}}(\mathbf{y}) = \frac{\mathbf{y} \cdot \mathbf{u}}{||\mathbf{y}||}\mathbf{u}$

c. $\text{proj}_{\mathbf{u}}(\mathbf{y}) = (\mathbf{y} \cdot \mathbf{u})\mathbf{u}$

d. $\text{proj}_{\mathbf{u}}(\mathbf{y}) = \mathbf{y} - \mathbf{u}$

The projection of $\mathbf{y}$ onto the line spanned by $\mathbf{u}$ is $\frac{\mathbf{y} \cdot \mathbf{u}}{||\mathbf{u}||^2}\mathbf{u} = \frac{\mathbf{y} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}}\mathbf{u}$.

31 [Intermediate] • Target: Orthogonal Projections

Compute the projection of $\mathbf{y} = [7, 6]$ onto $\mathbf{u} = [4, 2]$.

a. $[8, 4]$ ✓

b. $[4, 2]$

c. $[7, 6]$

d. $[2, 1]$

$\mathbf{y} \cdot \mathbf{u} = 7(4) + 6(2) = 28 + 12 = 40$. $\mathbf{u} \cdot \mathbf{u} = 4^2 + 2^2 = 20$. $\text{proj}_{\mathbf{u}}(\mathbf{y}) = \frac{40}{20}[4, 2] = 2[4, 2] = [8, 4]$.

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32 [Intermediate] • Target: Orthogonal Projections

If $\hat{\mathbf{y}} = \text{proj}_{\mathbf{u}}(\mathbf{y})$, what is the geometric relationship between $\mathbf{z} = \mathbf{y} - \hat{\mathbf{y}}$ and $\mathbf{u}$?

a. $\mathbf{z}$ is orthogonal to $\mathbf{u}$ ✓

b. $\mathbf{z}$ is parallel to $\mathbf{u}$

c. $\mathbf{z}$ has unit length

d. $\mathbf{z} = \mathbf{0}$ always

By the Orthogonal Decomposition Theorem, any vector can be uniquely decomposed into a component parallel to $\mathbf{u}$ and an orthogonal component $\mathbf{z} \perp \mathbf{u}$.

33 [Intermediate] • Target: Vector Arithmetic and Linear Combinations

Consider the vector space $\mathbb{R}^2$ with the standard dot product. Let $\mathbf{u} = (2, 3)$ and $\mathbf{v} = (-3, 2)$. Which of the following statements about $\mathbf{u}$ and $\mathbf{v}$ is true?

a. $\mathbf{u}$ and $\mathbf{v}$ are orthogonal. ✓

b. The angle between $\mathbf{u}$ and $\mathbf{v}$ is acute.

c. The angle between $\mathbf{u}$ and $\mathbf{v}$ is obtuse.

d. $\mathbf{u}$ and $\mathbf{v}$ are linearly dependent.

To determine if two vectors are orthogonal, we compute their dot product. If the dot product is zero, the vectors are orthogonal. For the given vectors $\mathbf{u} = (2, 3)$ and $\mathbf{v} = (-3, 2)$, the dot product is: $\mathbf{u} \cdot \mathbf{v} = (2)(-3) + (3)(2) = -6 + 6 = 0$. Since the dot product $\mathbf{u} \cdot \mathbf{v} = 0$, the vectors $\mathbf{u}$ and $\mathbf{v}$ are orthogonal. Therefore, the statement "$\mathbf{u}$ and $\mathbf{v}$ are orthogonal" is true. If the dot product were positive, the a.

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34 [Intermediate] • Target: Linear Independence and Dependence

Consider the matrix $A = (1, 1; 1, -1)$. Let $\mathbf{v} = (3; 1)$. Suppose we want to find the vector $\mathbf{x}$ of the form $c_1\mathbf{a}_1 + c_2\mathbf{a}_2$ where $\{\mathbf{a}_1, \mathbf{a}_2\}$ is an orthogonal basis derived from the columns of A, such that $||\mathbf{v} - \mathbf{x}||$ is minimized. Find the value of $||\mathbf{x}||$.

a. $\sqrt{10}$ ✓

b. $\sqrt{20}$

c. $\sqrt{5}$

d. $2\sqrt{10}$

The columns of matrix A are $\mathbf{c}_1 = (1; 1)$ and $\mathbf{c}_2 = (1; -1)$. This set of vectors is already orthogonal since $\mathbf{c}_1 \cdot \mathbf{c}_2 = (1)(1) + (1)(-1) = 0$. Thus, $\{\mathbf{a}_1, \mathbf{a}_2\} = \{\mathbf{c}_1, \mathbf{c}_2\}$ is an orthogonal basis. The problem asks to find the projection of $\mathbf{v}$ onto the subspace spanned by $\mathbf{a}_1$ and $\mathbf{a}_2$, which is the entire $\mathbb{R}^2$ in this case, as $\{\mathbf{a}_1, \mathbf{a}_2\}$ forms a.

35 [Intermediate] • Target: Dot Product, Norm, and Orthogonality

Consider the linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^3$ defined by $T(\mathbf{x}) = A\mathbf{x}$, where $A = (1, 2, 3; 4, 5, 6; 7, 8, 9)$. Let V be the image of T, i.e., $V = \text{Im}(T)$. V is a subspace of $\mathbb{R}^3$. Find the dimension of the orthogonal complement of V, denoted by $V^\perp$.

a. **1** ✓

b. 2

c. 0

d. 3

The image of the linear transformation T is the column space of the matrix A, i.e., $V = \text{Col}(A)$. The dimension of the image of T is the rank of the matrix A, i.e., $\dim(V) = \text{rank}(A)$. We can find the rank of A by reducing it to row-echelon form:
$A = (1, 2, 3; 4, 5, 6; 7, 8, 9) \to (1, 2, 3; 0, -3, -6; 0, -6, -12) \to (1, 2, 3; 0, -3, -6; 0, 0, 0)$ There are two non-zero rows, so the rank of A is 2. Thus, $\dim(V) = 2$. ; ; We kn.

36 [Intermediate] • Target: Orthogonal Projections

Consider a vector $\mathbf{v} = (3\ k)$ in $\mathbb{R}^2$. If the magnitude of $\mathbf{v}$ is $\sqrt{13}$, find the value of k given that $k > 0$.

a. 1

b. **2** ✓

c. 3

d. 4

The magnitude of a vector $\mathbf{v} = (v_1\ v_2)$ in $\mathbb{R}^2$ is given by $v = v_1^2 + v_2^2$. In this case, $\mathbf{v} = (3\ k)$ and $v = 13$. Therefore, we have the equation: $\sqrt{3^2 + k^2} = \sqrt{13}$. Squaring both sides gives: $3^2 + k^2 = 13$. This simplifies to $9 + k^2 = 13$. Subtracting 9 from both sides yields $k^2 = 4$. Taking the square root of both sides, we get $k = \pm 2$. Since the problem states that $k > 0$, the correct value for k is 2.

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37 [Intermediate] • Target: Vector Arithmetic and Linear Combinations

Consider the vector space $\mathbb{R}^2$ with the standard dot product. Let $\mathbf{u} = (3, 4)$ and $\mathbf{v} = (2, -1)$. Calculate the dot product $\mathbf{u} \cdot \mathbf{v}$.

a. 10

b. **2** ✓

c. -2

d. 6

The dot product of two vectors $\mathbf{u} = (u_1, u_2)$ and $\mathbf{v} = (v_1, v_2)$ in $\mathbb{R}^2$ is defined as $\mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2$. Given $\mathbf{u} = (3, 4)$ and $\mathbf{v} = (2, -1)$, we have: $\mathbf{u} \cdot \mathbf{v} = (3)(2) + (4)(-1)$ $\mathbf{u} \cdot \mathbf{v} = 6 - 4$ $\mathbf{u} \cdot \mathbf{v} = 2$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Linear Independence and Dependence • Cross-Lesson: 4.01 Determinants

Consider the vectors $\mathbf{u} = (2, -3)$ and $\mathbf{v} = (4, 1)$ in $\mathbb{R}^2$. What is the dot product $\mathbf{u} \cdot \mathbf{v}$?

a. 5

b. 8

c. **4** ✓

d. 11

The dot product of two vectors $\mathbf{u} = (u_1, u_2)$ and $\mathbf{v} = (v_1, v_2)$ in $\mathbb{R}^2$ is given by $\mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2$. For the given vectors, $\mathbf{u} = (2, -3)$ and $\mathbf{v} = (4, 1)$, the dot product is calculated as:
$\mathbf{u} \cdot \mathbf{v} = (2)(4) + (-3)(1) = 8 - 3 = 5$.

39 [Challenging] • Target: Dot Product, Norm, and Orthogonality • Cross-Lesson: 2.01 Systems of Linear Equations

Consider the matrix $A = (1, 2; -2, k)$. If the column vectors of A are orthogonal, find the value of k.

a. **2** ✓

b. -1

c. 4

d. -2

Let the column vectors of matrix A be $\mathbf{v}_1 = (1; -2)$ and $\mathbf{v}_2 = (2; k)$. For the column vectors to be orthogonal, their dot product must be zero. $\mathbf{v}_1 \cdot \mathbf{v}_2 = 0$ $(1)(2) + (-2)(k) = 0$ $2 - 2k = 0$ $2k = 2$ $k = 1$.

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40 [Challenging] • Target: Orthogonal Projections • Cross-Lesson: 3.01 Matrices & Matrix Algebra

Which of the following is a subspace?

a. x ✓

b. $\{p \in P_2 : p(0) = 1\}$

c. x

d. $\{A \in M_{2\times 2} : \det A = 0\}$

x is a subspace: a line through the origin, closed under addition and scaling. $\{p \in P_2 : p(0) = 1\}$: the zero polynomial is not in it. x: it does not contain the zero vector. $\{A \in M_{2\times 2} : \det A = 0\}$: not closed under addition: diag(1,0)+diag(0,1)=I.

41 [Challenging] • Target: Vector Arithmetic and Linear Combinations • Cross-Lesson: 4.01 Determinants

Consider the vector $\mathbf{v} = (5;\ -12)$. Calculate the magnitude of $\mathbf{v}$, denoted as x.

a. **13** ✓

b. 17

c. 7

d. 169

The magnitude of a vector $\mathbf{v} = (x; y)$ in $\mathbb{R}^2$ is given by the formula $= x^2 + y^2$. For the given vector $\mathbf{v} = (5;\ -12)$, we have $x = 5$ and $y = -12$. Therefore, $= (5)^2 + (-12)^2 = 25 + 144 = 169 = 13$.

42 [Challenging] • Target: Linear Independence and Dependence • Cross-Lesson: 2.01 Systems of Linear Equations

Which of the following is NOT a subspace?

a. x

b. $\{A \in M_{2\times 2} : A^T = A\}$

c. x ✓

d. x

x is not a subspace: it does not contain the zero vector. x: a line through the origin. $\{A \in M_{2\times 2} : A^T = A\}$: sums and multiples of symmetric matrices are symmetric. x: a line through the origin, closed under addition and scaling.

43 [Challenging] • Target: Dot Product, Norm, and Orthogonality • Cross-Lesson: 3.01 Matrices & Matrix Algebra

Consider the vector $\mathbf{v} = (5\ \ -12)$. Calculate the magnitude of $\mathbf{v}$, denoted as x.

a. **13** ✓

b. 7

c. 169

d. 17

$\mathbf{v} = (5\ \ -12)$. The magnitude of a vector $\mathbf{v} = (x\ y)$ in $\mathbb{R}^2$ is given by the formula $= x^2 + y^2$. Substituting the components of $\mathbf{v}$: $= 5^2 + (-12)^2 = 25 + 144 = 169 = 13$.

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44 [Challenging] • Target: Orthogonal Projections • Cross-Lesson: 4.01 Determinants

For which k is $\langle k, -\frac{1}{2}, \frac{3}{2}\rangle$ in $\text{span}\{\langle -1, 1, -3\rangle\}$?

a. $\frac{3}{2}$

b. $\frac{1}{2}$ ✓

c. $-\frac{1}{2}$

d. -1

The span of one vector is its multiples. Comparing the known entries gives the multiplier $-\frac{1}{2}$, so $k = -\frac{1}{2} \cdot (-1) = \frac{1}{2}$.

45 [Challenging] • Target: Vector Arithmetic and Linear Combinations • Cross-Lesson: 2.01 Systems of Linear Equations

Let B be the basis $\langle -1, -1, 1\rangle, \langle -1, -2, 2\rangle, \langle -2, 1, -2\rangle$ of R^3. Find $[v]_B$ for $\mathbf{v} = \langle 8, 3, -1\rangle$.

a. $\langle 8, 3, -1\rangle$

b. $\langle 2, 1, 9\rangle$

c. $\langle -3, -1, -2\rangle$ ✓

d. $\langle 3, 1, 2\rangle$

Solve $c_1\mathbf{b}_1 + c_2\mathbf{b}_2 + c_3\mathbf{b}_3 = \mathbf{v}$ (i.e. $B\mathbf{c} = \mathbf{v}$); the solution is $\langle -3, -1, -2\rangle$.

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